BTS –VI(SS) –01.16 – 0046 Reg. No. | | | | | | | | |

B

B. Tech. Degree VI Semester Special Supplementary Examination January 2016

CS 1603 OPERATING SYSTEM

(*2012 Scheme*)

Time: 3 Hours Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Explain the structure of an operating system.
(b) Explain briefly methods of inter process communication.
(c) Explain memory management using linked list.
(d) Explain concept of virtual memory.
(e) Explain RAID.
(f) Briefly explain (i) Clock (ii) Terminal.
(g) Define deadlock and condition for deadlock to occur.
(h) What is meant by starvation?

PART B

(4×15 = 60)

II. (a) Define mutual exclusion. Explain Petersons solution for mutual exclusion. (5)
(b) Explain : (i) Round Robin scheduling. (10)
(ii)First Come First Serve. Give examples for each.

OR

III. All four process arrive at time zero in the order given, while the length of the CPU burst time given in milliseconds. Consider FCFS, SJF, ROUND ROBIN (Quantum = 10 ms). Calculate the waiting time and find out which algorithm would give the minimum average waiting time. (15)

Process	Burst time
P1	10
P2	29
P3	3
P4	12

IV. What is page replacement algorithm? Explain any four page replacement algorithm with example. (15)

OR

V. Explain: (3 × 5 = 15)
(i) Translation Look Aside Buffer
(ii) Thrashing
(iii) Swapping.

VI. Explain any four disk scheduling algorithm with example. (15)

OR

VII. Explain: (3 × 5 = 15)
(i) Programmed I/O.
(ii) I/O Buffering.
(iii) Device drivers.

VIII. (a) Explain deadlock detection and recovery in detail. (8)
(b) Explain two phase locking protocol. (7)

OR

IX. Explain bankers algorithm in detail with example. (15)

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